Work History

Basic Information

Name: Sou Yoshihara (　)

Service	Account
GitHub	@sousquared
Instagram	@sousquared
Twitter	@sou_squared
Zenn	@sousquared

Personal Career Philosophy

Since childhood, I have loved drawing manga and illustrations so much that I once aspired to become a manga artist. With this background, I work with the desire to contribute to the creative and entertainment fields. My goal is to create products and works that bring joy to users.

Skills

Languages

- Japanese
  - Native
- English
  - Conversational
  - TOEIC 875 (Apr 2019)
  - TOEFL 85 (Jun 2017)

Programming Languages & Frameworks

- Python
- FastAPI
- Triton Inference Server
- JavaScript/TypeScript
- React
- Next.js
- NestJS
- Terraform

Database

- PostgreSQL

Cloud/Infrastructure

- GCP (Google Cloud Platform)
  - Cloud Run
  - Cloud Build
  - Cloud SQL
  - GKE (Google Kubernetes Engine)
  - Cloud Composer
  - etc.
- GitHub Actions

Development Experience

- PoC and production implementation using machine learning models
- API development (FastAPI)
- Clean Architecture and DDD (Domain-Driven Design) oriented coding

Others

- KaiRA (Kyoto University AI Research Association) Operating Member
  - Organizing reading sessions on artificial intelligence
- JEES SoftBank AI Talent Development Scholarship (2020-2021)

Strengths

- Can handle everything from PoC to production development
- Proactively taking action based on my interests
- Strong presentation and explanation skills
- International network of friends around the world

Interests

- Social applications of AI technology
- Mechanisms of human cognitive abilities (neuroscience, psychology, behavioral economics)
- AI-powered art and entertainment

Work Experience

Career Summary

As an engineer, I have not only developed features but also considered which PoCs and new features would become competitive advantages for the service and contribute to KPI improvements such as revenue growth, working backwards from business strategies. In the New Business Development team, I worked on improving product logic to reduce designers' production work, API development, and model serving. Also, as a leader of an ML team of 4-5 members, I led strategic planning and research roadmaps for updating internal product search functionality and new features, achieving quantitative results such as doubling search feature usage rates. In the Auto Generation team, I participated in 4 auto-generation logic projects in about 9 months, achieving service-level success rates in 3 of them. For example, I achieved 90% + quality in banner ad image auto-generation for a certain pattern within approximately 2 months. I built high-speed experiment pipelines and annotation-based improvement workflows, balancing development speed with quality improvement. I am currently the development lead for an internal annotation tool, expanding the tool's reach by deploying it to an overseas region and supporting confidential data. Also, as an operating member of DSOps training (2023–2025, serving as operating leader in 2024), I built a system where participating in training leads to actual projects in the field, contributing to organizational growth.

Aug 2018 - Aug 2020

Organization: Hitachi, Ltd. / Kyoto University Lab
Position: Student Researcher (Part-time)
Responsibilities

- Research and implementation of the latest machine learning papers
- Research and development of algorithms adaptable to modeling geospatial information
- Literature surveys on Graph Convolutional Networks and Relational Graph Convolutional Networks
- Implementation of parts of models devised by researchers
- Algorithm evaluation

Jan 2021

Organization: CyberAgent / Kiwami Prediction AI Division / Prediction Team
Position: ML/DS Intern
Responsibilities: Responsible for updating and accuracy evaluation of video ad score prediction models.

Apr 2022 ~ Mar 2025

Organization: CyberAgent / Kiwami AI Division / New Business Development Team
Position: ML Engineer
Responsibilities

I worked on improving product logic to reduce designers' production work, API development, and model serving.

Strategic Planning

- As a leader of the ML team, led strategic planning and roadmap creation for updating internal product search functionality and new features

Search Feature Improvement

- Formulated hypotheses from user usage history and improved functionality. Specifically, by adding multimodal search using images and text (e.g., searching with a material image + "premium feel"), improved the usability of image search and **successfully doubled search feature usage rates**. Also experimented and implemented logic to filter out poor-quality search results, **filtering out 40% of poor-quality search results** and improving the quality of search results displayed to users

Backend Development

- Also handled API development, mainly gaining experience in microservice development using Cloud Run

MLOps

- Under senior MLOps members, worked on building and operating basic inference servers and inference pipelines (using GKE, Triton, Cloud Batch, Apache Airflow on Cloud Composer, etc.). As a personal achievement, migrated the inference infrastructure from CPU to GPU. As a result, **achieved 6.28x faster inference speed compared to CPU**. Also reduced processing time by **62%** through preprocessing acceleration (including cache utilization)

Cost Optimization

- Cleaned up unused GKE and VMs, **contributing to approximately $2,500 monthly cost reduction**

Organizational Contribution

- As a leader of an ML team of 4-5 members, promoted PoC and production development of new features utilizing AI
- Also, as operating leader of DSOps training (2024), planned new initiatives and built a system where participating in training leads to actual projects in the field (see article for details. It is written in Japanese.)

Apr 2025 ~ Dec 2025

Organization: CyberAgent / Kiwami AI Division / Auto Generation Team
Position: ML Engineer
Responsibilities

Ad Auto-Generation Logic Development

- Participated in 4 auto-generation logic projects in about 9 months, achieving service-level success rates in 3 of them. Specifically, these projects involved consolidating ad patterns proposed by creators into

requirements, translating them to implementation level, and confirming success rates through human annotation. Collaborated with 1-2 engineers and 1-2 PMs, promoting speed-focused development

Project Results

- (Since specific logic details are confidential, I will describe based on success rates) Among the successful projects, in the 1st project, collaborated with 5 engineers and **achieved 74% success rate quality in auto-generation within 2 months** (this was quite high accuracy before NanoBananaPro was released)
- In the 2nd project, worked with 2 engineers on banner ad image auto-generation for a certain pattern and **successfully developed auto-generation logic with 90% success rate quality within 3 months**. In this project, I mainly handled prompt tuning for image generation and image filtering, and implementation of generation pipelines
- In the 3rd project, worked with 2 engineers on banner ad image auto-generation for a certain pattern and **successfully developed auto-generation logic with 90%+ quality within 2 months**. In this project, I handled prompt tuning for image filtering and clear verbalization of annotation criteria. In all projects, collaborated with ad creators on requirements definition, rapidly iterated through experiment implementation, annotation requests, and feature improvement PDCA cycles, conducting speed-focused development. Also handled API implementation when necessary

Organizational Contribution

- As an operating member of DSOps training (2025), realized a special lecture by Yuta Saito (Ph.D., Cornell University, Hanjuku Virtual Co., Ltd.). (I was the moderator for this article.)

Jan 2026 ~ Present

Organization: CyberAgent / AI Creative Company
Position: ML Engineer (Development Lead, Internal Platform Tool)
Responsibilities

As the development lead for an internal annotation tool (web application), I am responsible end to end, from requirements definition through production release and operations.

- **Expanding Usage**: Built a production environment in an overseas region and drove it through the first release, **making the tool available to a new group of users**. In addition to building the infrastructure, I led the fixes for issues found in the pre-release vulnerability assessment, and also worked with the legal team to prepare terms of use and a privacy policy for that region, and prepared manuals for user departments
- **Supporting Confidential Data**: Led project-level access control and role management from requirements definition through release, **making it possible to annotate highly confidential internal data**
- **Continuous Improvement**: Continuously released new task formats and usability improvements based on requests from user departments
- **Development Process**: Set up a development workflow using AI coding agents along with continuous deployment, **so that development and releases do not depend on any specific person**
- Main technologies: TypeScript (Next.js / NestJS), PostgreSQL, Terraform, GitHub Actions, GCP (Cloud Run / Cloud SQL)

Side Work / Freelance

Jul 2026 ~ Present

Organization: canuu (Side Work / Freelance)
Position: Engineer (AI-OCR Feature, from PoC to Production)
Responsibilities

For a sales management SaaS for seafood intermediate wholesalers, I was responsible for an AI-OCR feature that reads paper purchase slips from photos to assist data entry, from PoC through its production launch.

Problem Identification

- Built an accuracy evaluation framework (approximately 100 slip photos, human-verified ground-truth data, and automated scoring) and measured where errors occurred at each stage
- As a result, clarified that character recognition (OCR) on the slips was already at a practical level, and that the main cause of lower accuracy was **matching the item names written on the slips to the product codes managed by the wholesaler**
  - Recognition: amounts were correct on all evaluated rows, and about 80% of product names were read correctly
  - Matching: in practice, the same product is written differently by each supplier, and on the record side, multiple products are sometimes registered together under a generic code. For more than half of the ground-truth data, the correct code could not be determined from the item name alone
- Based on this analysis, proposed shifting the direction for improving accuracy from "finding a better OCR model" to "a mechanism that learns matching from human corrections"

Model Selection

- Compared multiple multimodal LLMs available on AWS Bedrock side by side under the same conditions using the evaluation framework above, and chose the model to adopt based on accuracy, cost, and output stability

Production Feature Improvements

- Changed the item-name-to-product mapping table so that it is built from human selections and updated when a selection is changed
- Recorded both the machine's answers and human corrections so that accuracy can be measured continuously
- Also published a trial environment (Docker, AWS App Runner) and prepared the configuration and documentation required for the production launch

Technologies Used: Python, Go, TypeScript / Next.js, AWS (Bedrock, App Runner), Anthropic API, Terraform, Docker, Claude Code

Research

- Sou Yoshihara, Taiki Fukiage, Shin'ya Nishida, "Towards acquisition of shape bias: Training convolutional neural networks with blurred images.", VSS, Poster session, 2021.
- Sou Yoshihara, Taiki Fukiage, Shin'ya Nishida, "Does training with blurred images bring convolutional neural networks closer to humans with respect to robust object recognition and internal representations?", Front. Psychol., Vol. 14,2023
- (Japanese) Sou Yoshihara, Taiki Fukiage, Shin'ya Nishida, "Shape Bias ", VISION, Vol. 33, No.1, 1-5, 2021. Best Presentation Award, Vision Society of Japan 2020 Summer Conference code

Articles

- " " —— 7 DSOps (Japanese)
- ML/DS | CyberAgent Way (Japanese) (I was the moderator for this article.)
- Codex MCP AI Coding : Codex 3 (Japanese)
- I also write mainly technical articles on Zenn: Zenn:@sousquared (Japanese)

Personal Projects

Vocavisual: Linking words with visuals, beyond your native language.

This project aims to directly connect images with words being learned, without relying on one's native language. For example, when learning the word "cat," instead of translating it to " " (cat in Japanese) and

memorizing it, the goal is to associate it with an image of a cat. By directly connecting images with words being learned without using one's native language as an intermediary, I believe this can promote deeper understanding. The project is being conducted on the following Instagram account:

- Vocavisual Korean: https://www.instagram.com/vocavisual_korean/